

Math 241, F14
Quiz 1

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (6 points) Find $3\mathbf{a} + 2\mathbf{b}$, $|\mathbf{a}|$ and $|\mathbf{a} - \mathbf{b}|$ with the vectors $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ and $\mathbf{b} = -2\mathbf{i} + 2\mathbf{j} + 9\mathbf{k}$.

$$\begin{aligned} 3\mathbf{a} + 2\mathbf{b} &= 3(\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}) + 2(-2\mathbf{i} + 2\mathbf{j} + 9\mathbf{k}) \\ &= (3 - 4)\mathbf{i} + (6 + 4)\mathbf{j} + (-9 + 18)\mathbf{k} \\ &= -\mathbf{i} + 10\mathbf{j} + 9\mathbf{k} \end{aligned}$$

$$|\mathbf{a}| = \sqrt{1^2 + 2^2 + (-3)^2} = \sqrt{14}$$

$$\mathbf{a} - \mathbf{b} = 3\mathbf{i} + 0\mathbf{j} - 12\mathbf{k} = 3\mathbf{i} - 12\mathbf{k}$$

$$\Rightarrow |\mathbf{a} - \mathbf{b}| = \sqrt{3^2 + (-12)^2} = \sqrt{153} = 3\sqrt{17}$$

Problem 2. (4 points) Find a unit vector that has the same direction as $\langle -2, 6, 3 \rangle$.

$$\begin{aligned} |\langle -2, 6, 3 \rangle| &= \sqrt{(-2)^2 + 6^2 + 3^2} \\ &= \sqrt{4 + 36 + 9} = \sqrt{49} = 7 \end{aligned}$$

$$\begin{aligned} \Rightarrow \mathbf{u} &= \frac{\langle -2, 6, 3 \rangle}{7} \\ &= \left\langle -\frac{2}{7}, \frac{6}{7}, \frac{3}{7} \right\rangle \end{aligned}$$